Rayen Ammar

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Profile

Third-year Artificial Intelligence and Data Science engineering student with hands-on experience designing AI, data, and full-stack applications across workforce analytics, recruitment, finance, ocean forecasting, and agentic RAG. Interested in engineering internships focused on AI engineering, data science, and intelligent applications.

Education

Engineering Degree in Artificial Intelligence & Data Science 2024–Present
TEK-UP University, Tunisia Coursework: Machine Learning, Statistics, Optimization, Databases, Generative AI

Experience

Full-Stack AI Developer Intern — Vermeg 2026
CHRONOS: Employee Time, Cost & Allocation Intelligence Platform | React/Vite, Spring Boot, Keycloak, PostgreSQL

- Designed a workforce intelligence platform tracking employee allocation and producing three-month cost forecasts using OLS regression with 95% confidence bands across companies and organisational units.
- Built anomaly detection for four operational risk types and implemented Keycloak OAuth2 role-based access for Admin, Finance Officer, and Product Manager dashboards, including dual CSV export.

Selected Projects

MedGuard AI — Agentic Drug Interaction & Medication Safety System Agentic AI | Advanced RAG / GraphRAG

- Developing an evidence-grounded medication-safety agent that evaluates drug interactions, contraindications, and medication risks in patient context, including age, kidney disease, and pregnancy.
- Designing a LangGraph agent workflow with tool calling and evidence-verification routing; combining a Neo4j drug knowledge graph with Qdrant vector retrieval and openFDA/RxNorm APIs for drug identification and evidence retrieval.

Stack: Python, LangChain, LangGraph, GraphRAG, Neo4j, Qdrant, openFDA API, RxNorm API

AI-Powered Dual-Lane Customer Support System RAG | NLP | MLOps

- Designed an AI-assisted support architecture for internal IT and external customer requests, combining ticket classification and routing, RAG-grounded response suggestions with citations, and ticket-volume anomaly alerts.
- Specified a human-in-the-loop workflow and MLOps lifecycle covering experiment tracking, model registry, CI/CD, drift monitoring, and controlled retraining/rollback.

Stack: Python, Hugging Face Transformers, scikit-learn, LangChain, LLMs, vector database, FastAPI, React, MLflow, Docker, GitHub Actions, Evidently, Grafana

CHRONOS — Employee Cost & Allocation Intelligence Platform React | Spring Boot | PostgreSQL

- Built workforce allocation analytics, three-month cost forecasting, anomaly detection, role-based dashboards, and CSV export workflows.

DUALIS — AI Financial Lending Platform .NET | ML.NET | SQL Server | GenAI

- Built a predictive model for loan eligibility and risk from applicant financial profiles; integrated generative AI to explain lending decisions in human-readable language.

UNIVIOO — AI CV & Recruitment Platform Django | NLP | React

- Implemented candidate–opportunity matching with TF-IDF and cosine similarity, plus an LLM-powered chatbot for personalised career guidance.

waveAI — Ocean Intelligence Platform Python | PyTorch | Scikit-learn

- Built ML/deep-learning models for storm detection and ocean forecasting; used SHAP to explain predictions from satellite and oceanographic data.

Retail Store Sales Forecasting LightGBM | XGBoost | Time Series

- Engineered calendar and temporal features, trained models with TimeSeriesSplit, and visualised seasonal and store-level patterns using Plotly and Seaborn.

Technical Skills

Programming: Python, SQL, C#, JavaScript, Java **ML/DL:** Scikit-learn, PyTorch, ML.NET, LightGBM, XGBoost, NLP, time-series forecasting
GenAI/RAG: LangChain, LangGraph, Advanced RAG, GraphRAG, LLM APIs, TF-IDF, embeddings, tool calling, evidence verification
Data/Graph: Pandas, NumPy, Matplotlib, Plotly, Power BI, PostgreSQL, SQL Server, Neo4j, Qdrant
Frameworks/Cloud: Django, Spring Boot, ASP.NET Core, React, Streamlit, FastAPI, REST APIs
MLOps/Tools: MLflow, Docker, Git, DVC, CI/CD, GitHub Actions, Evidently, Grafana, Keycloak

Certifications & Activities

Certifications: Microsoft Azure AI Fundamentals; NVIDIA Generative AI with Diffusion Models; NVIDIA Deep Learning Fundamentals; Prompt Engineering (Udemy).
Languages: Arabic (Native), English (Fluent), French (Fluent), German (Beginner).
Activities: Digital Future 6.0 Conference — Data Literacy; participant, IEEE GODS Competition Hackathon.